Scaling Laws for Generative Reward Models

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Abstract

We study generative reward models (GenRMs) as online preference providers for policy optimization in creative writing. Across Qwen3 reward-model sizes from 0.6B to 14B, Thinking judges trained with GRPO have higher static Gold-label accuracy than Answer-Only judges trained with SFT. This ranking often reverses when the judges train policies through online DPO: Answer-Only rewarders yield higher peak and top-five-average policy Elo in seven of eight matched settings under a fixed, learned 32B Gold evaluator. Intermediate Thinking-judge checkpoints can also outperform later, more statically accurate checkpoints as rewarders. Sweeps over rewarder size, policy size, training duration and DPO regularization show substantial benefits from reward-model capacity, including judges larger than the policy. We analyze trajectories with evaluator-specific Elo arenas, conditional match-resampling intervals and descriptive scaling fits. These results demonstrate an *evaluator–rewarder gap* in the studied pipelines: fixed-distribution judging accuracy does not determine downstream optimization quality. The comparison changes judge-training objective as well as response format, and the main ranking is conditional on its learned reference rather than a universal human-quality result. A limited follow-up using a different learned judge is inconclusive overall.

1 Introduction

Policies trained via reinforcement learning from human feedback systematically exploit imperfections in their reward models, degrading true quality even as proxy metrics improve [Gao et al., 2022, Rafailov et al., 2024]. Generative reward models (GenRMs) have emerged as a promising response: by formulating evaluation as conditional text generation rather than scalar prediction, GenRMs can leverage pre-training priors and chain-of-thought reasoning and achieve strong performance on static preference benchmarks [Mahan et al., 2024, Zhang et al., 2025, Whitehouse et al., 2025]. This success has fostered an implicit assumption that more accurate static evaluators should translate to more effective online rewarders.

Prior work has already shown that static reward-model accuracy can be a poor predictor of downstream optimization [Wen et al., 2025, Frick et al., 2024, Kim et al., 2025]. We study this distinction specifically for generative rewarders as the policy distribution changes, targeting a practitioner-facing question: *if one chooses a GenRM as the reward signal, how should compute and architecture be allocated to maximize downstream policy quality?* We study three design dimensions in the online setting where they matter: (i) thinking versus answer-only judging, (ii) judge scale, and (iii) policy scale. We instantiate these axes in creative writing using LitBench [Fein et al., 2025], a non-verifiable preference domain where correctness cannot be programmatically checked. In such domains, static evaluator accuracy can saturate while reward hacking and

Observation	Evidence in this paper
Static evaluation favors Thinking GenRMs	Thinking judges have higher Gold-label validation accuracy across the studied sizes (Figure 1).
Online rewarder quality is different	In online DPO, Answer-Only judges usually give higher peak and top-five Gold Elo than Thinking judges in the matched sweep (Table 3, Figure 2).
Judge scale matters most reliably	Larger Answer-Only GenRMs improve policy quality even when the judge is much larger than the 4B policy; Thinking judge scale helps but saturates earlier (Figure 4).
Static accuracy can be over-trained for reward use	Intermediate Thinking checkpoints can outperform fully-trained Thinking checkpoints as online rewarders (Figure 2b).

Table 1: A compact map of the paper's central evidence. The main empirical distinction is between GenRMs as *evaluators* on a fixed distribution and GenRMs as *rewarders* inside an online optimization loop.

proxy-Gold divergence remain salient, making creative writing an ideal testbed for whether stronger static evaluators translate to better in-loop reward signals.

Contributions. (1) We present a systematic scaling study of GenRMs deployed as online reward signals, sweeping judge format, judge size (0.6B–14B), policy size (0.6B–8B) and training budgets under online DPO. (2) We trace policy trajectories under both proxy and Gold evaluators, using within-arena comparisons to separate rising proxy scores from gains under the reference evaluator. (3) We provide empirical guidance within the studied pipelines: judge scale and training duration matter strongly, Answer-Only rewarders usually sustain higher Gold-rated quality than Thinking variants, and intermediate GenRM checkpoints can outperform later models as rewarders.

Our findings reveal an *evaluator–rewarder gap*. Thinking GenRMs have higher static validation accuracy across the studied sizes, yet this ranking often reverses during online optimization. Answer-Only GenRMs usually yield higher peak Gold Elo, with exceptions where Thinking remains competitive. Additional judge capacity remains valuable above policy size; this does not establish that judging is intrinsically harder than generating.

2 Methodology

We begin with a human preference dataset of the form

$$\mathcal{D}_{\text{human}} = \{x^{(i)}, y_A^{(i)}, y_B^{(i)}, I_H^{(i)}\}_{i=1}^{N},$$

where $x^{(i)} \in \mathcal{X}$ are prompts, $y_A^{(i)}, y_B^{(i)} \in \mathcal{Y}$ are pairs of responses to the prompts, and $I_H^{(i)} \in \{A, B\}$ denotes the human-preferred response.

Gold evaluator. Since human preferences are not available on demand during policy training, we train a large generative reward model, denoted V_{gold}, on $\mathcal{D}_{\text{human}}$. This frozen, queryable evaluator supplies the reference preferences for the experiment. Human supervision grounds that reference on the static data distribution; it does not guarantee human agreement on later policy-generated responses. We therefore study the downstream behavior of smaller proxy GenRMs *relative to this learned anchor*, not an independently established measure of human writing quality.

To make evaluation computationally feasible, we choose V_{gold} to be an *Answer-Only* model that outputs a single indicator token. Using V_{gold}, we construct a Gold Preference dataset:

$$\mathcal{D}_{\text{gold}} = \{x^{(i)}, y_A^{(i)}, y_B^{(i)}, I_G^{(i)}\}_{i=1}^{M},$$

which is then used to train a variety of smaller GenRMs.

Proxy GenRMs. We consider two types of GenRMs: *Answer-Only* and *Thinking*. The Answer-Only models output a direct judgment token (A or B):

$$I \sim v_{\text{ans}}(\cdot \mid x, y_A, y_B),$$

while the Thinking models first generate a reasoning trace z before producing the final verdict:

$$z \sim v_{\text{think}}(\cdot \mid x, y_A, y_B), \quad I \sim v_{\text{think}}(\cdot \mid x, y_A, y_B, z).$$

Training objectives. We train Answer-Only models using supervised fine-tuning (SFT) and Thinking models using GRPO. SFT directly supervises the verdict token; GRPO rewards final-verdict accuracy and positional consistency without requiring a reference reasoning trace. This compares two practical training pipelines, rather than isolating the causal effect of reasoning under identical objectives. The 4B Thinking ablation below provides evidence for the chosen GRPO recipe, not an exhaustive comparison of training methods.

For GRPO, we employ two reward signals:

1. **Accuracy reward:** A binary reward (1 if the model reaches the correct verdict, 0 otherwise).
2. **Positional consistency reward r_{pos}:** Inspired by prior work, we observe that models often produce contradictory judgments when the order of (y_A, y_B) is swapped in the prompt. The reward manager links both orderings using a shared pair identifier and computes a majority vote within each ordering's sampled completions. An additional reward of 1 is assigned to correct completions only when both majorities are correct and consistent after normalizing response order. This cross-order reward aggregation is distinct from GRPO advantage normalization, whose groups are keyed by the sampled prompt row.

We refer to these trained GenRMs as *Proxy models*, and evaluate them with respect to V_{gold} preferences.

Policy optimization. We train policies π_θ of varying sizes using *Online Direct Preference Optimization* (Online DPO). DPO is a natural choice for preference-based policy optimization because GenRMs produce pairwise preferences as their native output, which DPO can consume directly without requiring scalarization or additional transformations. In *offline* DPO, response pairs are sampled once from a fixed reference policy and reused across many gradient updates, keeping the supervision distribution static throughout training. In *online* DPO, response pairs are repeatedly resampled from the current policy after each update, so the supervision distribution evolves as the policy changes. This online setting is particularly important for our study, as we aim to understand GenRM robustness under distribution shift induced by policy optimization. Empirically, online preference optimization has been shown to yield more stable and effective alignment dynamics when implemented carefully [Guo et al., 2024].

Formally, at online update t, we sample $x \sim \mathcal{P}_{\text{train}}$ and two responses $y_1, y_2 \sim \pi_{\theta_t}(\cdot \mid x)$. A proxy GenRM v orders the pair as (y_v^+, y_v^-), where y_v^+ is preferred to y_v^-. The DPO loss induced by this proxy is

$$\mathcal{L}_{\text{DPO}}(\theta; v, \pi_{\theta_t}) = -\mathbb{E}\left[\log \sigma\left(\beta\left[\log \frac{\pi_\theta(y_v^+ \mid x)}{\pi_{\text{ref}}(y_v^+ \mid x)} - \log \frac{\pi_\theta(y_v^- \mid x)}{\pi_{\text{ref}}(y_v^- \mid x)}\right]\right)\right].$$

The idealized Gold-aligned update replaces v with V_{gold}. Where the proxy disagrees with Gold matters as well as how often it disagrees: different response pairs contribute different gradient

directions and magnitudes. Verdict filtering also changes which sampled pairs reach the update. Static agreement on $\mathcal{D}_{\text{gold}}$ measures neither effect on the evolving pair distribution. Appendix B.3 gives a short DPO interpretation, without claiming a measured gradient-level explanation of the results.

The policies are trained on prompts sampled from the same broad creative-writing source distribution as the Gold and Proxy datasets, with LitBench prompts excluded by the intended construction procedure. We periodically sample responses on a fixed validation set and estimate Elo ratings from pairwise comparisons under the proxy and Gold evaluators. Each evaluator defines a separate fitted arena: numerical equality between its Elo and another evaluator's Elo is not response-level agreement. Appendix B.1 describes the actual Bradley–Terry fits, checkpoint summaries and conditional uncertainty.

The empirical question is whether a higher-accuracy judge on the fixed Gold-labeled validation set also trains a higher-rated policy under Gold. Agreement on human-story pairs and agreement on evolving policy-generated pairs are different measurements. Our experiments compare the static ranking with downstream policy outcomes; they do not reconstruct trustworthy response-level online agreement from aggregate Elo ratings.

3 Experimental Setup

Dataset construction. Our experimental design separates human supervision, Gold-derived supervision and policy-generation prompts:

- **Human preference dataset $\mathcal{D}_{\text{human}}$:** We sample 22,000 preference triplets (x, y_A, y_B, I_H) from LitBench Fein et al. [2025], a large-scale preference dataset over human-written stories from Reddit. We split these into 21,000 for training and 1,000 for validation. This dataset supplies human supervision for the Gold evaluator.

- **Gold preference dataset $\mathcal{D}_{\text{gold}}$:** We sample a separate, non-overlapping set of 22,000 triplets from LitBench and re-annotate them using the trained Gold evaluator, yielding (x, y_A, y_B, I_G) triplets. While both datasets are sourced from LitBench, the former reflects human judgments and the latter reflects the Gold model's judgments. The intended split is 21,000 for training and 1,000 for validation, before filtering invalid or inconsistent Gold labels. This dataset is used for training and evaluating Proxy GenRMs.

- **Policy training dataset:** We source prompts from the WritingPrompts dataset[1], the ancestor dataset from which LitBench was constructed. To avoid data leakage, we exclude all prompts that appear in LitBench, yielding 199,000 candidate prompts. We randomly sample 91,000 prompts (90,000 for training and 1,000 for validation) unless specified otherwise in specific experiments.

Building on prior work that has extensively evaluated various reward models on LitBench facilitates future research and cross-comparisons between our results and other studies.

Model families. We use the Qwen3 model family for all GenRMs and policies. Qwen3 provides both Thinking and Answer-Only modes across multiple parameter sizes, enabling controlled comparisons where we can vary GenRM mode and scale while holding the backbone architecture and instruction-tuning distribution fixed. For policy models, using the same backbone as the GenRMs allows us to cleanly isolate the interaction between policy size and GenRM size, which is central to our scaling analysis.

[1] `https://huggingface.co/datasets/euclaise/WritingPrompts_preferences`

Gold GenRM training. We train the Gold evaluator from QWEN3-32B using supervised fine-tuning on $\mathcal{D}_{\text{human}}$. To improve positional robustness, we train on both orderings (y_A, y_B) and (y_B, y_A), resulting in 42,000 ordered training examples before any input filtering. We use a batch size of 128 and a learning rate of 1×10^{-5}.

Proxy GenRM training. We train Proxy GenRMs from the QWEN3 series at sizes {0.6B, 1.7B, 4B, 8B, 14B} using $\mathcal{D}_{\text{gold}}$. We refer to off-the-shelf models as `Baseline-[size]-Ans` and `Baseline-[size]-Think`. Answer-Only models use supervised fine-tuning with batch size 128 and learning rate 1×10^{-5}. Both positional orders are used, giving up to 42,000 ordered examples before Gold-label filtering. Thinking models use GRPO with 16 prompts per step, 8 completions per prompt, configured minibatch size 64 and learning rate 1×10^{-6}. The retained launcher disables the trainer-level KL reward controller but enables a separate actor KL loss with coefficient 3×10^{-5}; this is not a zero-KL setup. We refer to trained proxies as `GenRM-[size]-Ans` and `GenRM-[size]-Think`.

Policy training. We train policies π_θ from the QWEN3 series at sizes {0.6B, 1.7B, 4B, 8B} using Online DPO with thinking disabled. We use a prompt batch size of 192, global minibatch size of 64 preference pairs, learning rate 1×10^{-6} and $\beta = 0.02$ unless varied. Two current-policy responses are generated per prompt. The proxy judges both orderings, and the trainer sums order-normalized verdicts, discarding zero-sum pairs. A malformed verdict contributes zero, so one valid verdict and one malformed verdict can still yield a retained pair; this is not a guarantee of two valid agreeing votes. The original experiment description specifies prompt and response caps of 512 and 2048 tokens, respectively. The retained hand-edited launcher instead contains caps of 500 and 1500; resolved per-run configurations have not been recovered for every plotted trajectory, so those original caps should be read as nominal settings rather than a newly verified run-wide invariant.

Evaluation. During policy training, we save validation generations every 10 training steps; the retained launcher saves model weights every 50 steps. We call the evaluated policy states evaluation snapshots, without implying that every state has separately retained weights. We compute Elo ratings from pairwise comparisons judged by the proxy and Gold evaluators. Each evaluator has its own fitted ratings over the relevant contestants and comparisons; axes from different evaluators are not one shared numerical scale. Main experiments run for about 460 steps, giving 47 evaluation snapshots including step zero; ablations can differ.

Artifact reconstruction. Dataset counts above follow the contemporaneous experiment account and retained construction code. They describe raw splits, not an independently reconstructed census of every consumed post-filtering example. Some original filtering and Gold-training-label preparation steps, and the complete per-run configuration ledger, are not retained in the inspected source. We distinguish these provenance limits from the stored generations, rating tables and figures used in the analysis.

Reproducibility and LLM usage. Beyond the GenRM and policy models described above, we did not use LLMs for dataset filtering, rubric creation, label consolidation, or prompt generation. All prompts used for policy generation and GenRM pairwise evaluation are provided in Appendix C.

4 Experiments and Results

4.1 GenRM Training

Gold evaluator validation. The Gold GenRM achieves 79% agreement with human preferences on the held-out validation split of $\mathcal{D}_{\text{human}}$. Prior work on LITBENCH reports peak accuracies of approximately 78% Fein et al. [2025]. This supports the Gold model as a useful reference on the

Axis	Values swept	Default / anchor
Proxy GenRM format	Answer-Only SFT, Thinking GRPO	Both compared
Proxy GenRM size	0.6B, 1.7B, 4B, 8B, 14B	4B unless varied
Policy size	0.6B, 1.7B, 4B, 8B	4B unless varied
Online DPO β	0.1, 0.02, 0.01, 0.005, 0.002	0.02
Training trajectory	Policy evaluation every 10 steps; intermediate Thinking GenRM checkpoints	About 460 policy steps

Table 2: Axes studied in matched and fixed-axis online-DPO sweeps, not a full factorial grid. The design separates fixed-distribution accuracy from in-loop rewarder behavior.

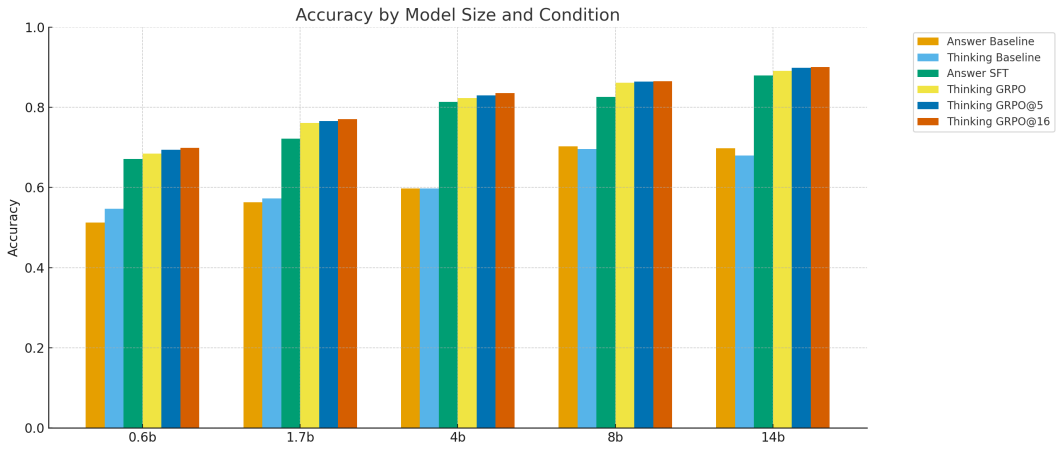


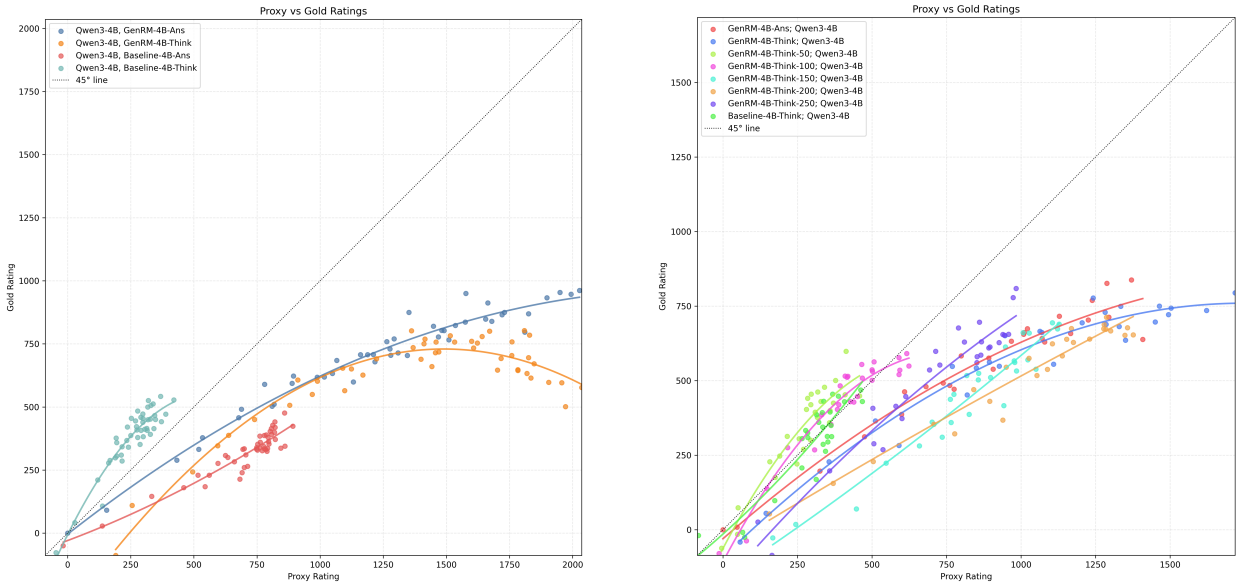
Figure 1: Static evaluator accuracy on the Gold Preference dataset. Thinking GenRMs exceed Answer-Only models across the studied sizes, while both trained variants substantially exceed off-the-shelf baselines. The size of the static advantage varies by model.

held-out static distribution; it does not measure its human agreement on policy-generated outputs after optimization.

Trained GenRMs substantially outperform baselines. Both Answer-Only SFT and Thinking GRPO improve static Gold-label accuracy over their off-the-shelf counterparts (Figure 1). Similar baseline accuracies do not rule out evaluator-specific preferences or format bias. The trained-versus-baseline improvement is substantial across the sizes studied.

GRPO is effective in the 4B Thinking ablation. On the Gold Preference dataset, the off-the-shelf model achieves 60.0%, STaR achieves 60.7%, distillation from Qwen3-32B thinking traces (SFT-style) achieves 64.7%, and GRPO with accuracy and positional consistency rewards achieves 83.2%. GRPO gives the largest improvement among these tested recipes, motivating its use for the Thinking proxies.

Thinking GenRMs outperform Answer-Only models on in-distribution evaluation. Thinking models have higher static Gold-label accuracy at each studied size (Figure 1); the gap varies by size. Majority voting over $k = 5$ and $k = 16$ Thinking completions gives modest further gains. Given the additional computation, we use one completion per judged ordering during policy training.



(a) Trained vs. baseline GenRMs

(b) Intermediate Thinking checkpoints

Figure 2: Proxy vs. Gold ELO ratings for policies trained with different GenRMs. (a) Comparison of trained vs. baseline GenRMs. (b) Policies trained with intermediate Thinking GenRM checkpoints. Axes show separately fitted Proxy and Gold Elo; the dotted diagonal marks equal numerical ratings, not identical verdicts. **Takeaway:** The evaluator-best checkpoint need not be the rewarder-best. The original experiment account specifies 40,000 training prompts for (b).

4.2 Trained GenRMs vs. Baseline Models in Policy Training

RESEARCH QUESTION. Does training the GenRM improve downstream policy quality compared to off-the-shelf models?

We compare policies optimized with trained versus off-the-shelf 4B GenRMs, with the policy also fixed at 4B (Figure 2a). Here "baseline" describes the rewarder, not an untrained policy. Trained rewarders produce higher peak Gold ratings in both modes. That is not uniform dominance at every training duration: at matched step 460, the off-the-shelf Thinking rewarder's policy reaches 542 Gold Elo versus 501 for the trained Thinking rewarder's policy, after the latter declines from a higher peak of 803. Answer-Only retains a trained-rewarder advantage throughout the matched trajectory, with peaks 961 versus 476. The restored chronological comparison in Appendix A makes the late crossover and the peak/final distinction explicit. While Thinking has the stronger off-the-shelf rewarder peak, Answer-Only has the stronger trained-rewarder peak in this arena.

4.3 Answer-Only vs Thinking

RESEARCH QUESTION. Which of the compared Answer-Only SFT and Thinking GRPO pipelines provides the better online rewarder?

In the shared global Gold arena, Answer-Only rewarders give higher peak policy Elo in 7 of 8 matched configurations (Table 3). The exception is the 0.6B policy with a 4B judge: Thinking peaks at 1048 versus 1013 for Answer-Only. Averaging each run's five highest-rated snapshots gives the same 7-of-8 pattern, with mean gaps of 107 Elo for top-five averages and 126 for peaks. Under 200 independent-row bootstrap resamples and regularized refits, 5 of 8 peak gaps and 5 of 8 top-five gaps

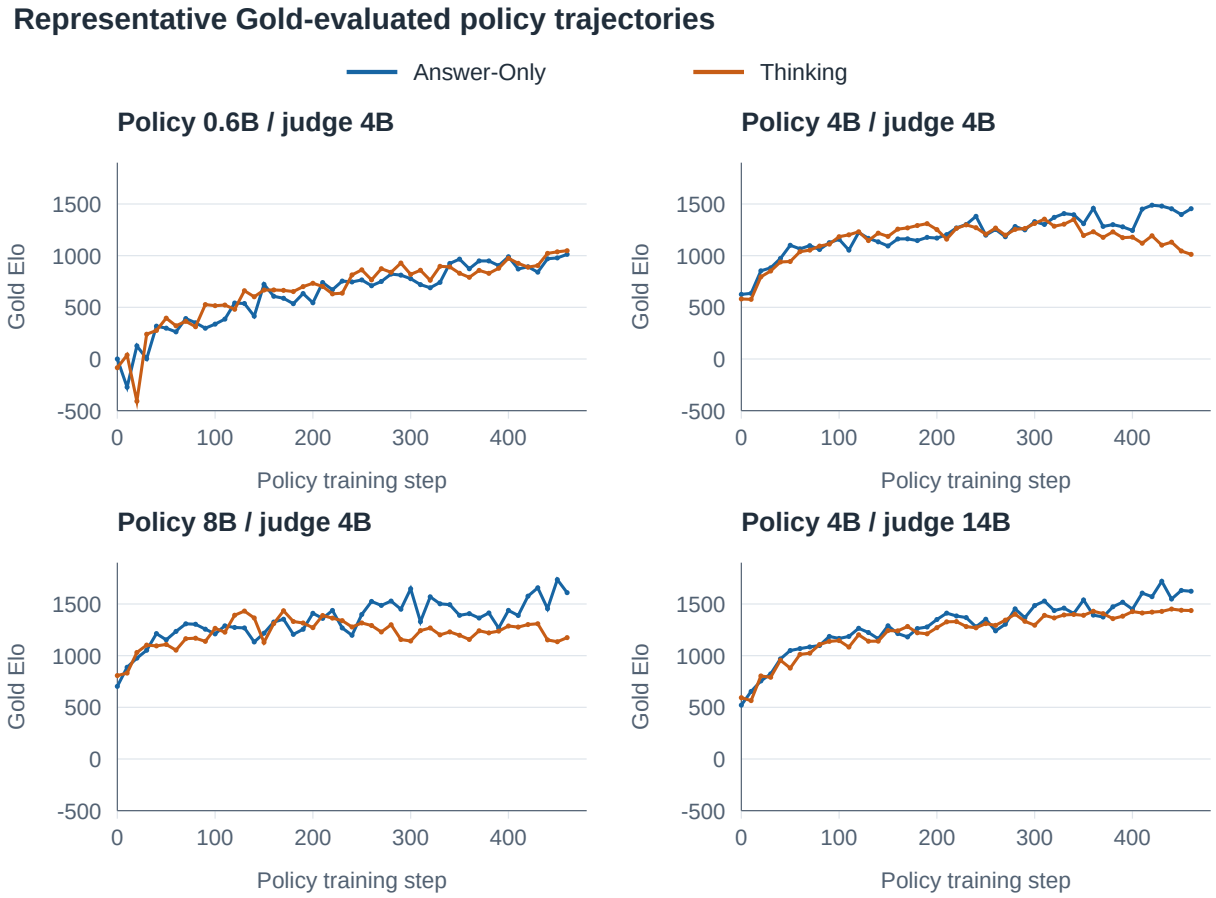


Figure 3: Four representative Gold-rated policy trajectories for Answer-Only SFT (blue) and Thinking GRPO rewarders (orange), including the smallest-policy exception, the 4B anchor and larger-policy/rewarder cases. All panels share the global Gold arena and vertical limits; lines connect recorded training snapshots, not fitted curves. All eight matched configurations are summarized in Table 3 and shown chronologically in Appendix A.

have 95% percentile intervals above zero. This is conditional match-table uncertainty, not prompt-cluster, generation or retraining uncertainty (Appendix B.1). Figure 3 shows representative Gold-rated training trajectories; Appendix A provides the complete matched set, including competitive Thinking settings. The reversal matters because static evaluation favors Thinking, and the off-the-shelf baseline comparison also differs from the trained comparison. Under the learned Gold reference, the statically stronger practical judge-training pipeline is not generally the better online reward provider. The difference is consistent with different responses to policy-induced distribution shift, but does not identify that mechanism: Gold's own preferences can also change with its inputs, and both judge training and response mode differ. Two observations help locate the gap without resolving its cause.

First, higher in-distribution GenRM accuracy does not necessarily translate to better policy training. Figure 2b compares intermediate 4B Thinking checkpoints at 0, 50, 100, 150, 200, 250 and 300 GRPO steps. The contemporaneous experiment account reports static accuracies of 59.7%, 61.5%, 64.5%, 68.7%, 71.5%, 79.2% and 82.3%, respectively. The 250-step judge (79.2%) produces a stronger online rewarder than the later, more statically accurate checkpoint in this ablation. This illustrates that evaluator-optimal and rewarder-optimal checkpoint selection can differ. This late

Policy	GenRM	Peak gap [95% CI]	Top-5 gap [95% CI]
0.6B	4B	−35 [−134, 49]	−19 [−59, 25]
1.7B	4B	+113 [43, 176]	+94 [48, 134]
4B	0.6B	+41 [−32, 104]	+24 [−2, 65]
4B	1.7B	+145 [104, 239]	+159 [133, 214]
4B	4B	+135 [49, 241]	+141 [93, 189]
4B	8B	+41 [−67, 143]	+25 [−18, 88]
4B	14B	+269 [128, 488]	+193 [134, 257]
8B	4B	+302 [158, 573]	+243 [189, 343]

Table 3: Matched Answer-Only versus Thinking Gold Elo gaps in the global all-configuration arena, reported as Answer-Only minus Thinking. Confidence intervals come from 200 bootstrap resamples of the stored Gold match table. Answer-Only wins 7 of 8 matched cells under both peak and top-5 checkpoint averaging; the exception is the smallest-policy setting with a 4B judge.

ablation used 40,000 policy-training prompts rather than the main 90,000.

Second, GRPO supervises final preferences rather than the correctness of each reasoning step. A Thinking GenRM may therefore learn traces that support training-distribution labels without remaining reliable under policy-induced shift; we treat this as a hypothesis, not a measured mechanism. The trajectory data show differing stability: across the eight matched global configurations, Answer-Only runs lose an average of 60 Elo from peak to final, compared with 130 for Thinking. An auxiliary verdict-agreement reconstruction could align rows only by checkpoint-player pairs and row order, not unique prompt/response identifiers. We do not use its percentages as verified response-level agreement or as mechanistic evidence.

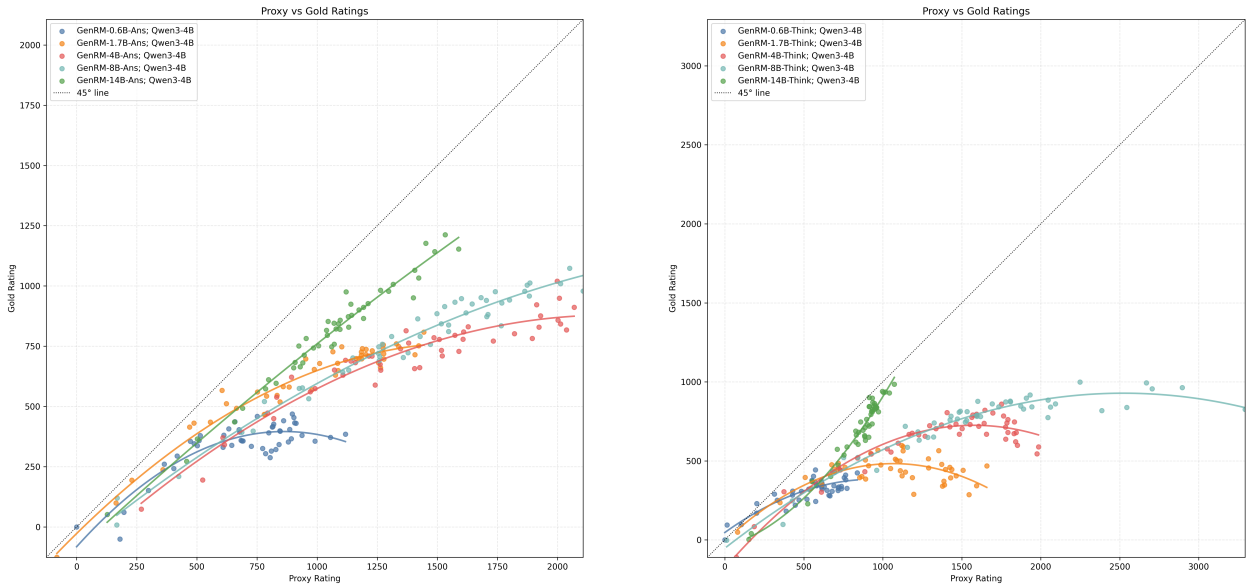
These results should not be read as evidence that Answer-Only supervision preserves stylistic diversity. A surface audit of stored peak-checkpoint generations in two 4B-policy cells shows that high Gold Elo can coexist with template convergence: in the 4B-policy/4B-judge cell, 94% of Answer-Only peak outputs share the same opening line, and the duplicate-line rate is higher than in the corresponding Thinking peak outputs. The stronger Answer-Only result is therefore a preference-optimization result, not a diversity result. In summary, Answer-Only rewarders require shorter verdict generations and, in these runs, usually sustain higher Gold-rated policy quality. This does not independently establish greater human-quality robustness or identify reasoning as the cause. Intermediate Thinking checkpoints remain useful candidates for rewarder selection.

4.4 Effect of GenRM Size

Research question. Does scaling the reward model improve policy quality, and do gains continue when the GenRM exceeds the policy in size?

We investigate the effect of scaling GenRM size while fixing the policy size at 4B. Figure 4 shows results for both Answer-Only and Thinking GenRMs across scales from 0.6B to 14B. We observe clear gains from increasing GenRM size, especially for Answer-Only rewarders. With a 4B policy, the Answer-Only 14B judge is the strongest judge-size configuration in the global arena. Thinking rewarders also benefit substantially from scale, but less monotonically: the 8B Thinking judge slightly exceeds the 14B Thinking judge in peak Gold Elo (1507 vs. 1451) and top-5 average Gold Elo (1488 vs. 1437). Thus additional rewarder capacity can remain useful above policy size, while its benefit depends on the judge's training and online behavior. Parameter counts alone do not establish a general ranking of the difficulty of judging and generating.

Appendix B.2 summarizes the existing sweep with descriptive scaling fits. A log-linear model



(a) Answer-Only GenRMs

(b) Thinking GenRMs

Figure 4: Proxy vs. Gold ELO ratings for policies trained with GenRMs of different sizes. The policy is fixed at 4B. Left: Answer-Only GenRMs (0.6B–14B). Right: Thinking GenRMs (0.6B–14B). The diagonal marks equal numerical ratings in separate arenas. **Takeaway:** Larger Answer-Only rewarders improve Gold Elo beyond policy size; Thinking scaling is less monotone.

Configuration	Top-5 avg Gold Elo	Peak Gold Elo	Peak step
GenRM-4B-Ans; Policy 8B	1645	1736	450
GenRM-14B-Ans; Policy 4B	1630	1719	430
GenRM-8B-Ans; Policy 4B	1513	1549	460
GenRM-8B-Think; Policy 4B	1488	1507	460
GenRM-4B-Ans; Policy 4B	1467	1488	420
GenRM-14B-Think; Policy 4B	1437	1451	440

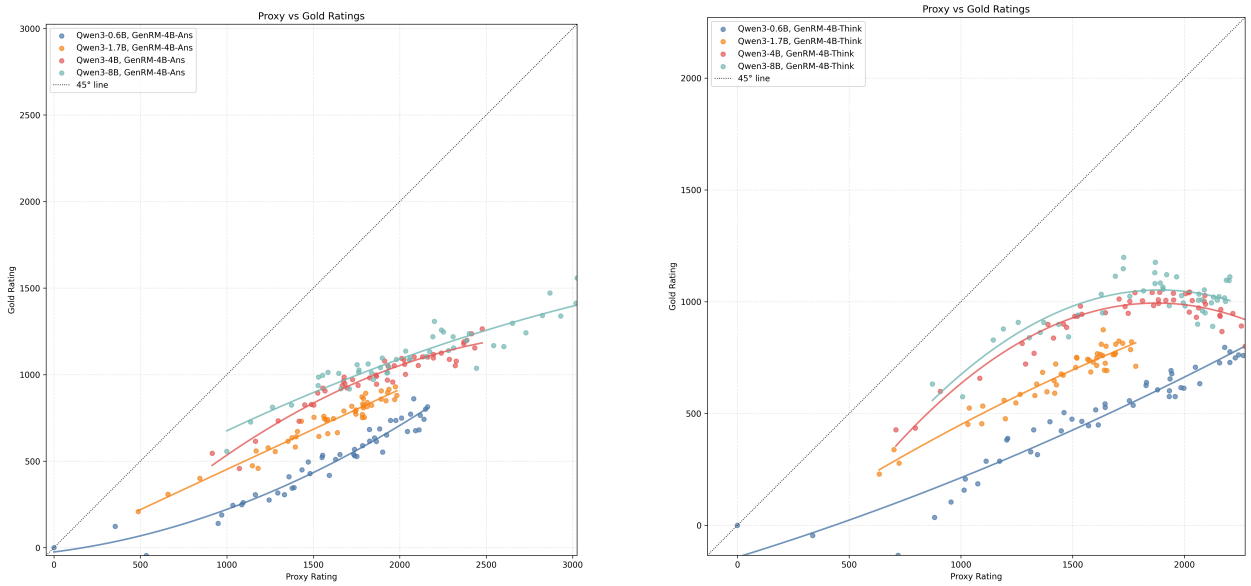
Table 4: Top online-DPO configurations in the global all-configuration arena, ranked by average Gold Elo over each run's best five checkpoints. The practical frontier is dominated by larger judges and Answer-Only rewarders, with the 8B Thinking judge remaining competitive.

using policy size, judge size and judge mode has an in-sample R^2 of 95.2% for top-5 Gold Elo across the 16 matched global configurations. In that fit, the coefficients correspond to 176 Elo per Answer-Only policy-size doubling and 136 per judge-size doubling; the Thinking-run values are 119 and 116 Elo. These are fitted summaries of this sparse grid, not predicted causal gains from a new doubling or a compute-optimal allocation rule.

4.5 Effect on Policy Size

RESEARCH QUESTION. Does scaling the policy improve quality, and does the effect differ between Answer-Only and Thinking supervision?

We examine the effect of scaling policy size while fixing the GenRM. Figure 5 shows results for policies ranging from 0.6B to 8B trained with both Answer-Only and Thinking GenRMs. For Answer-Only supervision, larger policies achieve higher Gold Elo across the studied scales. This



(a) Answer-Only GenRMs

(b) Thinking GenRMs

Figure 5: Proxy vs. Gold ELO ratings for policies of different sizes trained with fixed GenRMs. Left: Answer-Only GenRMs. Right: Thinking GenRMs. The diagonal marks equal numerical ratings in separate arenas. **Takeaway:** Larger policies improve under Answer-Only supervision in this sweep; large Thinking-supervised policies saturate earlier.

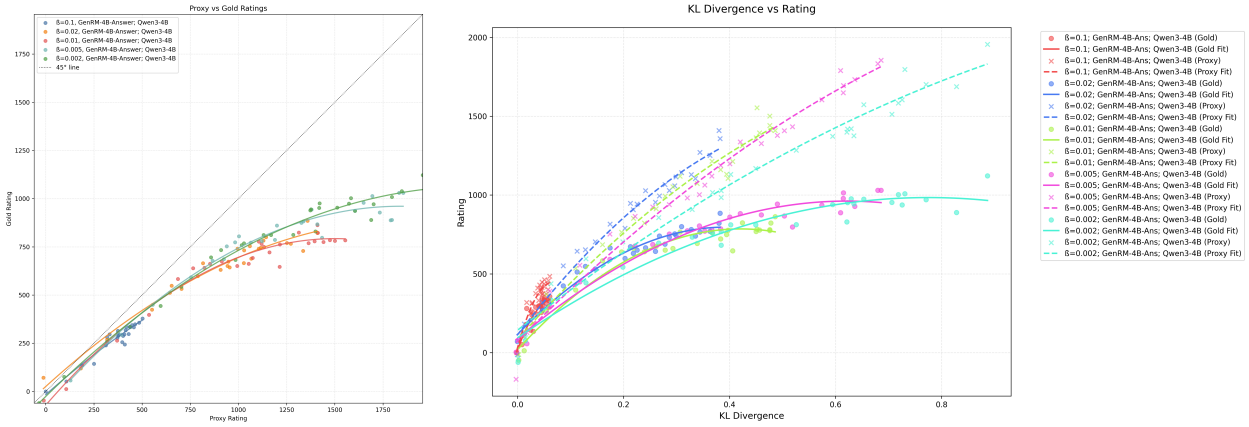
measures increased quality under the learned reference, not independent evidence that larger policies generalize better to human preferences.

For Thinking supervision, however, the picture is less clear. While the largest policy does achieve the highest peak performance, its Gold ELO curve saturates and bends downward earlier than smaller policies, which continue to improve steadily. In the policy-size arena, the peak checkpoint for Thinking supervision moves earlier as policy size increases: the 0.6B and 1.7B policies peak near the end of training, the 4B policy peaks around step 320, and the 8B policy peaks around step 130. This suggests two possible explanations: (i) large policies may more quickly exhaust the effective capacity of the GenRM, reaching its "ceiling" earlier, or (ii) beyond a certain scale, further increasing policy size without correspondingly stronger GenRMs may have diminishing or negative returns. Additional training is required to disentangle these explanations, but the evidence points toward an important asymmetry: scaling policy size is reliably beneficial under Answer-Only supervision, but under Thinking supervision, the interaction between policy capacity and GenRM capacity is more fragile.

4.6 Effect of the β Coefficient in Online DPO

RESEARCH QUESTION. How does the DPO regularization strength affect Gold ratings and measured policy KL?

We analyze the effect of the β coefficient in Online DPO. Figure 6(a) shows Proxy vs. Gold Elo for $\beta \in \{0.1, 0.02, 0.01, 0.005, 0.002\}$. The smallest tested value, $\beta = 0.002$, gives the highest Gold Elo. The ordering is not strictly monotone: 0.02 slightly exceeds 0.01 in the complete stored trajectory. Lower β does not simply increase Proxy ratings at Gold's expense: Gold performance also improves over the observed range. Proximity to the diagonal is a descriptive comparison of



(a) Proxy vs. Gold ratings across β values

(b) KL divergence vs. rating

Figure 6: Effect of the β coefficient in Online DPO. Left: Proxy vs. Gold ELO ratings for Answer-Only GenRMs at different β values. Right: measured KL divergence versus rating under different β values. The diagonal marks equal numerical ratings in separate arenas. **Takeaway:** Smaller β gives higher Gold Elo among tested values. The original experiment account specifies 40,000 training prompts for this sweep.

separately anchored ratings, not proof of exact proxy–Gold verdict agreement. Figure 6(b) shows the measured KL–Gold-rating tradeoff. Smaller tested β values permit greater policy movement and higher Gold ratings in this sweep; this does not establish a universal exploration rule or a safe divergence threshold. The result favors less regularization within the tested range, without identifying an optimum outside it. In the KL-matched plotted subset, peak Gold Elo rises from 357 at $\beta = 0.1$ to 1122 at $\beta = 0.002$, while the corresponding KL estimates rise from 0.061 to 0.887. The complete stored $\beta = 0.1$ trajectory reaches about 378 Elo at step 260, beyond that matched subset. KL values are empirical estimates; small near-zero negative estimates are not interpreted as negative true KL.

5 Limitations and Discussion

Gold evaluator is a learned proxy, not humans. We fine-tune a 32B Gold evaluator on human preferences, re-annotate pairwise data with this model, and discard items with inconsistent Gold verdicts [Gao et al., 2022]. This selects a more position-consistent label set but does not eliminate label error or the Gold model's inductive biases. Gold is larger than the proxy judges; its size and static human agreement do not guarantee reliable judgments throughout policy optimization. *Takeaway.* Interpret results as optimization toward a specified learned reference.

Evaluator dependence and response quality. Gold uses an Answer-Only Qwen3 model, and proxies share its model family and derived labels. Baseline comparisons do not rule out evaluator-specific preferences. A retained follow-up audit used a different learned judge, `Qwen3-235B-A22B-Instruct-2507`, on 40 randomly selected validation prompts from each of two 4B-policy settings, comparing each mode's peak-Gold outputs with randomized presentation order. Answer-Only won 42 of 80 comparisons (52.5%; 95% Wilson interval 41.7–63.1%): 17/40 with 4B rewarders and 25/40 with 14B rewarders. This limited audit is inconclusive overall and does not independently establish the Gold-ranked advantage. It uses another Qwen3 Answer-Only judge, a different rubric and Gold-selected checkpoints, not human evaluation or a cross-family replication. Appendix B.4 gives the procedure. *Takeaway.* The main result is conditional on the specified learned Gold

reference; neither format neutrality nor universal writing-quality superiority is established.

Distribution-dependent influence of judge fine-tuning. One unresolved hypothesis is that, on shifted policy outputs, fine-tuned judges increasingly share the preferences of their original models. If the learned Answer-Only Gold evaluator and Answer-Only proxies share such priors, part of the measured advantage could reflect reference alignment rather than human preference. The retained Elo trajectories do not test this mechanism. A discriminating check would compare Gold with its own unfine-tuned 32B parent on identical human-story and early/late policy response pairs, controlling response order and output validity. Agreement alone would still not establish human quality or its decline. We report this as a hypothesis, not a result or an explanation established by this study.

Domain scope. The trained rewarder and online-DPO sweeps use creative writing; we do not systematically evaluate safety, helpfulness or verifiable tasks such as math and code. A limited OpenOrca diagnostic uses only off-the-shelf models (Appendix B.5); it does not replicate the trained-rewarder comparison or policy optimization in another domain. *Takeaway.* The central result is scoped to these creative-writing pipelines; domain transfer remains open.

Relation to prior creative-writing studies. Creative-writing preferences are subjective. Optimizing win rate can reduce stylistic diversity. We observe some style convergence [Chung et al., 2025] in late-stage policies against both judge modes, while draw rates do not show large collapses. Fein et al. [2025] report that chain-of-thought can degrade verification accuracy for creative writing, and that trained BT and generative verifiers outperform zero-shot judges on their benchmark. In our setting, distilling Qwen3-32B traces into Qwen3-4B also yields negligible gains as evaluators, which aligns with these observations. However, after training, thinking judges significantly improve as static evaluators. The gap between thinking and answer-only during policy optimization therefore cannot be explained by domain "unsuitability" alone; it indicates different optimization dynamics. *Takeaway.* Our evaluation–optimization divergence is a property of the training loop in this domain, not only a property of static judging.

Family and algorithm scope, and behavioral priors. All models use Qwen3 backbones where pretraining data are not public. Cross-family studies suggest that behavioral priors, including synthetic data that instantiate verification or backtracking, can modulate RL improvements and collapse family gaps [Gandhi et al., 2025]. Such priors could shift our coefficients. We do not evaluate PPO-style RLHF for policies or alternative thinking-judge recipes. Additionally, we do not include Best-of-N (BoN) inference-time comparisons; our focus is on GenRMs as *training signals* for policy optimization rather than inference-time scaling, where the primary gains in our setting arise from gradient-based learning rather than reranking at generation time. BoN comparisons remain valuable future work for understanding the full compute-performance tradeoff. *Takeaway.* Our coefficients and inflection points are conditional on online DPO and Qwen3 underlying behavioral priors.

Elo anchoring and schedule. Ratings depend on evaluator, contestants, anchor and match schedule [Chiang et al., 2024]. Proxy and Gold ratings are fitted separately; the diagonal of a proxy–Gold plot is not a verdict-agreement statistic. We report global and size-stratified arenas. Bootstrap intervals for Answer-Only-minus-Thinking gaps independently resample stored winner/loser rows and refit a lightly regularized Bradley–Terry model. They are conditional on these stored comparisons, not robust to arbitrary dependence between rows sharing prompts or responses, and do not capture fresh generation, retraining, seed, anchor or tournament-design uncertainty. *Takeaway.* Interpret within-arena differences alongside these dependence and selection limits, not as universal confidence statements.

Post-hoc scaling-law fits. The parameters in Appendix B.2 are fitted to the stored sweep after the experiments were run. They are compact statistical summaries of the observed grid, not

seed-averaged laws or extrapolations to larger models. Most stored KL trajectories do not reach a full downturn region. *Takeaway.* Local slopes and curvature summarize this sweep; stronger laws require independent seeds and deliberately extended optimization budgets.

6 Related Work

Reward overoptimization is studied in RLHF [Ouyang et al., 2022, Gao et al., 2022] and direct methods such as DPO [Rafailov et al., 2023, 2024]. Prior evaluations explicitly distinguish static accuracy from downstream policy performance [Wen et al., 2025, Frick et al., 2024, Kim et al., 2025]; the generic distinction is not new to this work. GenRMs replace scalar-head prediction with text generation, allowing verdicts and rationales [Mahan et al., 2024, Zhang et al., 2025]. Subsequent approaches include GRPO and positional consistency in J1 [Whitehouse et al., 2025], self-principled critique and aggregation in DeepSeek-GRM [Liu et al., 2025], and test-time majority voting [Wang et al., 2025, Shi and Jin, 2025]. Related supervision uses generated critiques [Yu et al., 2025], criteria trees [Liang et al., 2025] and reasoning plans [Saha et al., 2025]; judge reliability and static benchmarks remain active topics [Ye et al., 2024, Lambert et al., 2024]. Our contribution is the measured set of generative-rewarder capacity and training-pipeline sweeps inside online DPO, using evaluator-specific Elo arenas [Chiang et al., 2024]. The results concern practical Answer-Only SFT and Thinking GRPO pipelines under a shared model family, not a causal isolation of reasoning format.

7 Conclusion

We study generative reward models as online training signals. Under our learned Gold reference, Thinking GenRMs outperform Answer-Only variants on static evaluation, yet this advantage usually reverses during policy optimization. The result concerns the compared SFT/GRPO pipelines in this domain, not an isolated causal effect of reasoning or a universal ranking of writing quality.

Three findings offer practical guidance within these experiments: (1) reward model scale is a major driver of Gold-evaluated policy outcomes, with gains continuing even when the GenRM exceeds the policy in size; (2) intermediate GenRM checkpoints can outperform fully-trained models as rewarders; and (3) in-distribution accuracy does not by itself predict training effectiveness. Selecting GenRMs only by static benchmarks can therefore select checkpoints that are suboptimal for downstream optimization under the learned reference.

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A Retained Empirical Trajectories and Capacity Sweeps

These figures restore the recorded trajectories behind the aggregate results. They use retained CSVs only: no new model calls, fitted curves or inferred observations. Snapshots are not independent training replications.

A.1 Trained and Off-the-Shelf Rewarders over Training

Figure 7 separates the common Gold evaluation from the four rewarders' own-proxy ratings.

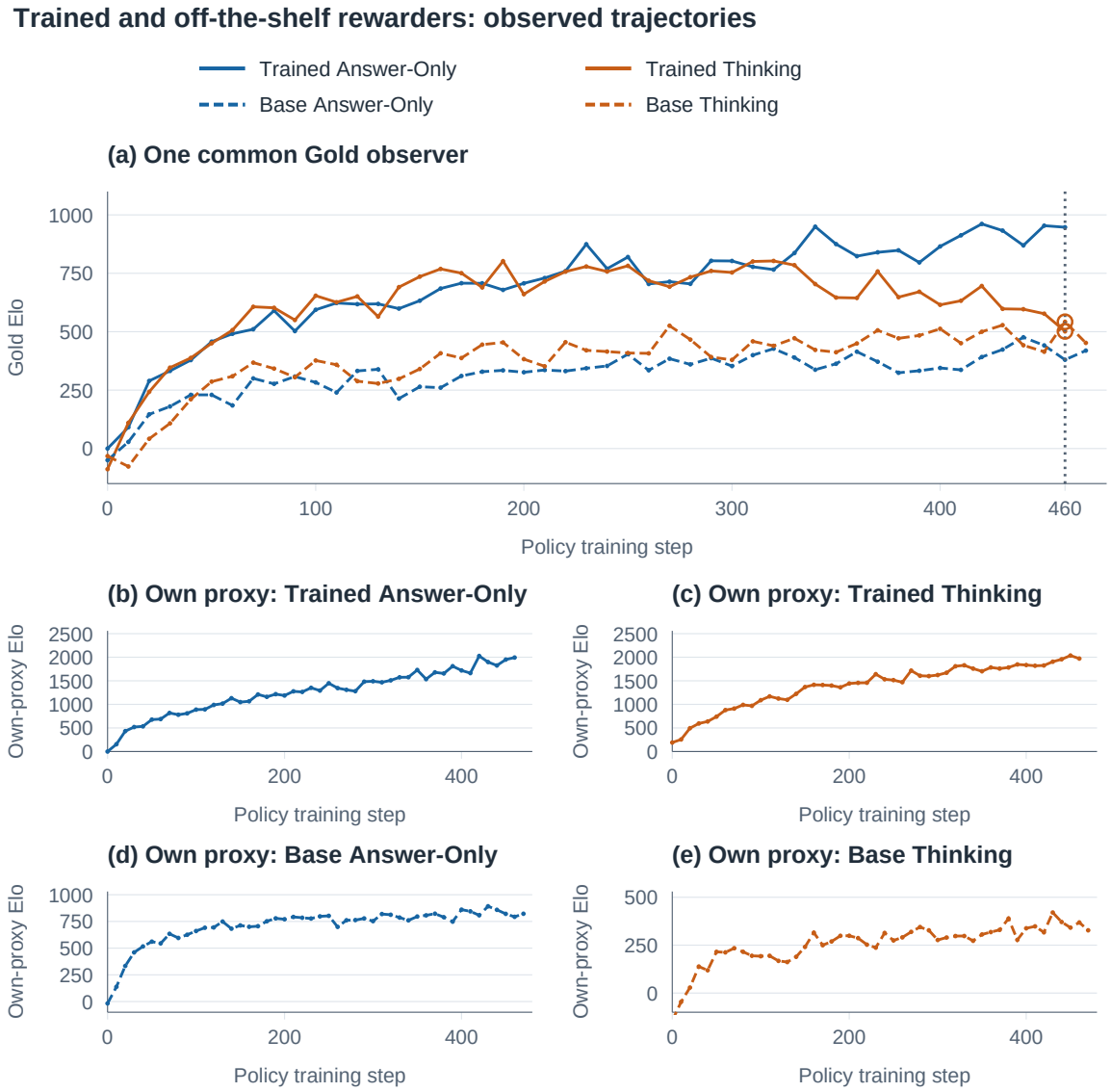


Figure 7: **Trained versus off-the-shelf rewarders through the retained training horizon.** Blue denotes Answer-Only and orange Thinking; solid lines denote trained rewarders and dashed lines off-the-shelf rewarders. The upper panel uses one Gold arena; outlined points mark step 460. Each lower panel uses its own rewarder's separately fitted ratings, so heights across those panels are not comparable. Policy and judge are 4B. Source: `trained_vs_baseline/*.csv`; no smoothing or missing-step interpolation.

Trained rewarders achieve higher Gold peaks in both formats. For Thinking, the trained peak is 802.8 versus 542.3, but at shared step 460 the baseline leads 542.3 to 501.5. At 450 the ordering is opposite (414.3 versus 577.1). Only the baseline continues to 470. This late crossover is not a significance result or uniform baseline superiority. From 320 to 460, the trained Thinking policy's own-proxy rating rises 1808.3 to 1971.5 while Gold falls 802.8 to 501.5: proxy–reference divergence, not an identified mechanism or human-quality decline.

A.2 All Eight Matched Conditions in One Gold Arena

Figures 8 and 9 show the complete stored Gold-rated trajectories for all eight matched conditions. All panels use `all_params/gold.csv`, sharing one arena and vertical scale. They expose early gains and late reversals hidden by best-checkpoint summaries; a falling tail does not itself measure verdict agreement or Gold-to-base reversion.



Figure 8: **Policy-size sweep with a fixed 4B rewarder.** All four panels use the common Gold arena, common vertical limits and observed policy-training steps. Blue: Answer-Only SFT rewarders; orange: Thinking GRPO rewarders. Each line represents one retained policy trajectory, not a mean across seeds. Source: `all_params/gold.csv`.

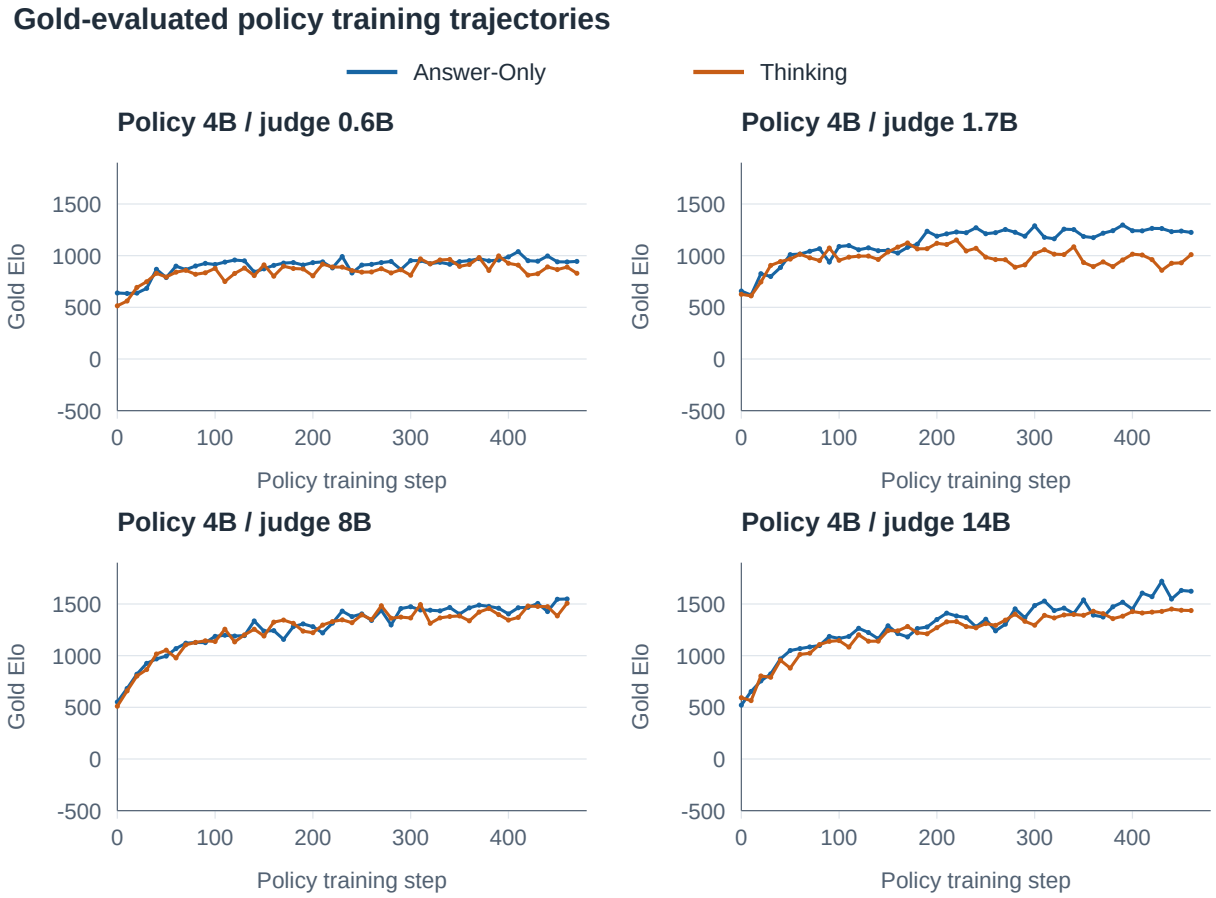


Figure 9: **Additional rewarder-size conditions with a fixed 4B policy.** Together with the 4B/4B panel of Figure 8, these are the five rewarder sizes studied. Colors, arena and axis limits match the preceding figure. The retained curves show measured behavior, not a claim of monotonic improvement at every step or across every rewarder size.

A.3 The Same Gold Observations against Recorded KL Estimates

Figures 10 and 11 provide a complementary view using the retained measured policy-KL exports rather than training step. The exports do not contain explicit checkpoint IDs. For each exported Gold row, we require a *unique* match on its exact trajectory prefix and Gold rating (absolute tolerance 10^{-7}) in the common-arena file. This recovers 752 measured rows. Four additional global-arena snapshots have no matching KL observation and are omitted, not assigned an estimated KL. The machine-readable join ledger records every source row and recovered checkpoint.

The horizontal coordinate is the recorded estimator, not an inferred value from a scaling fit. Small negative finite-sample estimates near the initial policy are retained rather than clipped. Lines follow checkpoint order even where the estimated KL moves backwards. Neither this estimate nor policy step is a validated distance from the Gold evaluator's human-story training distribution.

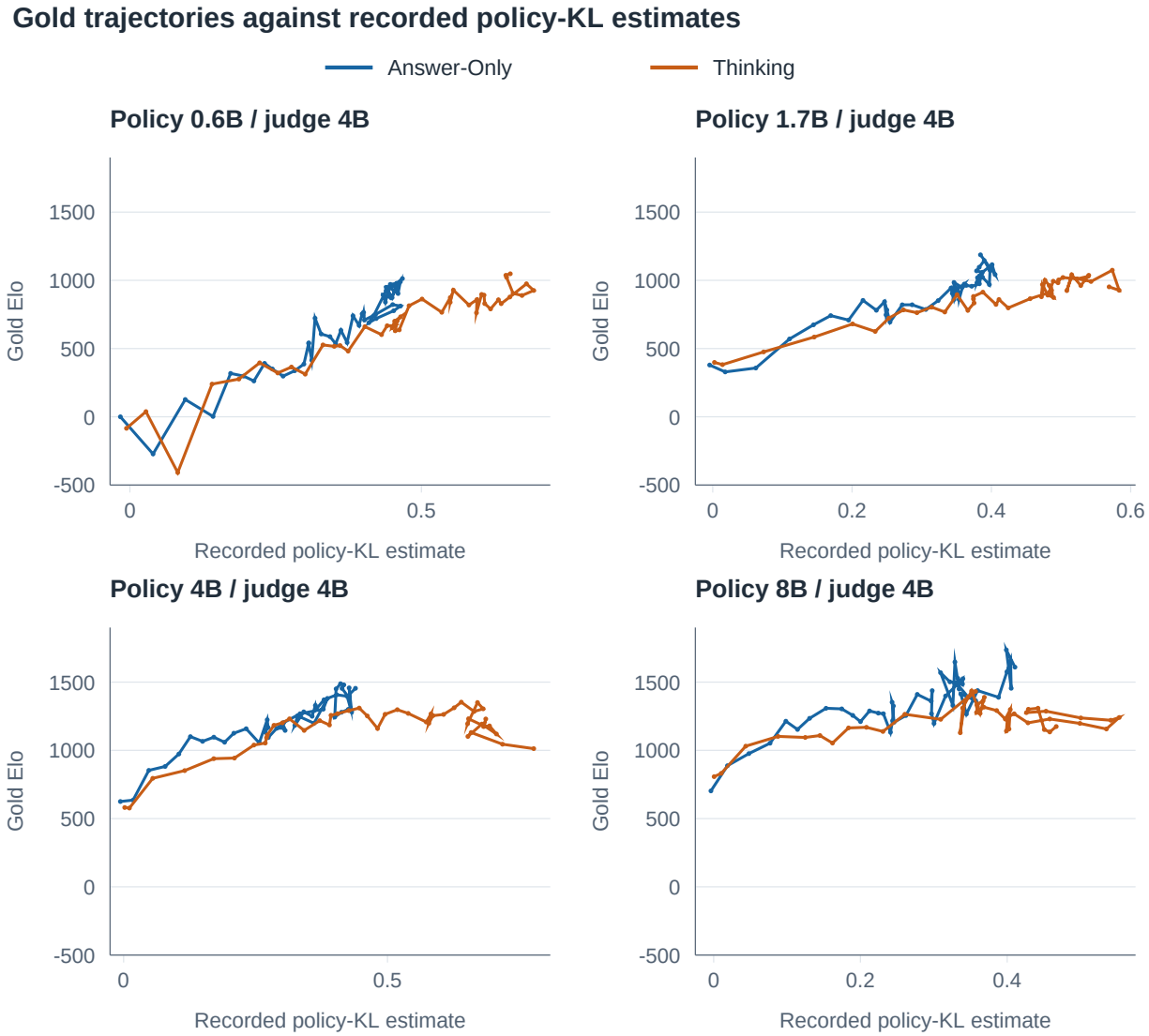


Figure 10: **Policy-size conditions against recorded KL estimates.** Gold coordinates are the same common-arena observations as Figure 8; only the horizontal coordinate changes. Lines follow the recovered checkpoint order. Each cell has 47 measured snapshots per mode; the two extra continuation snapshots in the 1.7B-policy cell lack measured KL and are excluded. Horizontal limits differ across cells; vertical limits and the Gold arena are shared.

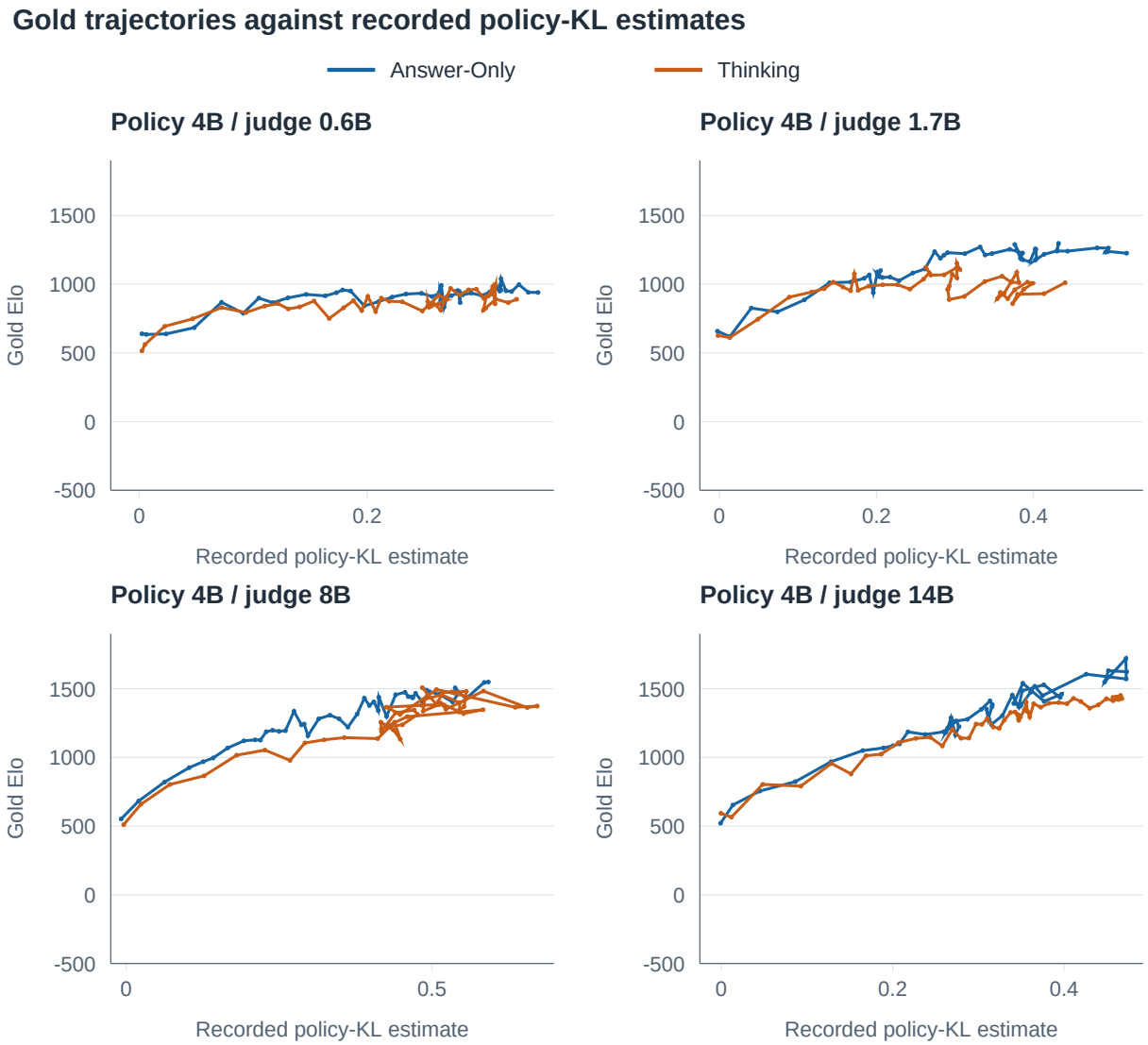


Figure 11: **Rewarder-size conditions against recorded KL estimates.** Gold coordinates match Figure 9. Only uniquely joined measured KL observations are shown. The two extra continuation snapshots in the 0.6B-rewarder cell are absent from these KL exports and are excluded. Small negative estimates remain visible. This is an empirical view, not a fitted Gold-versus-KL scaling law.

A.4 Capacity Allocation without Sorting Away the Experimental Axes

Figure 12 preserves both sweep directions: rewarder size at fixed policy size, and policy size at fixed rewarder size. The mean of the five highest Gold-rated snapshots is still selected under Gold, not an independently validated deployment rule. Only observed cells are shown.

Even parameter-count-matched swaps need not favor the same allocation in both formats. At a 12B summed parameter count, the 8B-policy/4B-rewarder condition has Answer-Only top-five Gold Elo 1645.0, versus 1513.1 for the 4B-policy/8B-rewarder condition. Thinking reverses that ordering (1402.2 versus 1488.1). A parameter sum is not an inference- or training-compute budget, especially when one rewarder generates reasoning traces; these observed swaps do not identify a universal compute-optimal allocation.

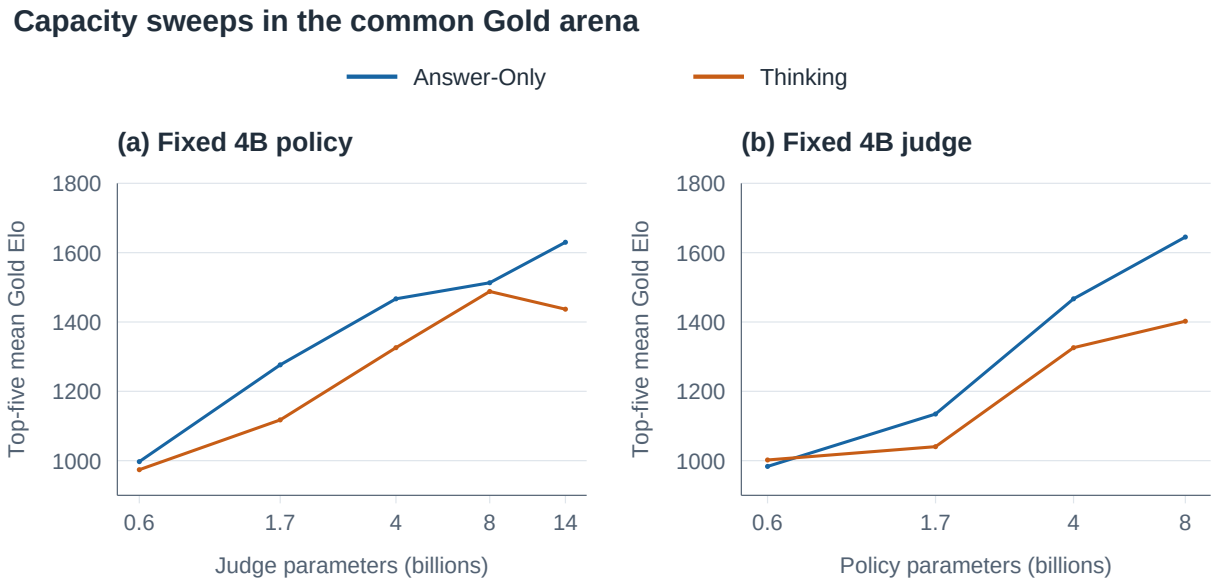


Figure 12: **Observed capacity sweeps in the common Gold arena.** Left: rewarder size varies while the policy is 4B. Right: policy size varies while the rewarder is 4B. Each point is the average of the five highest recorded Gold-rated snapshots of that trajectory. Parameter-size axes are logarithmic and labeled with actual sizes. Blue: Answer-Only; orange: Thinking. These are descriptive observed summaries, not compute-normalized optima, predictive scaling fits, or independent-seed confidence intervals.

Intermediate judge training. Figure 2b retains the intermediate-Thinking-checkpoint ablation. The original account specifies judge steps 0, 50, 100, 150, 200, 250 and 300 and a 40,000-policy-prompt budget. We preserve that image rather than infer static accuracies or checkpoint identities from pixels; its source account does not substitute for an exact per-judge run manifest.

Reconstruction and provenance. The generator `scripts/build_empirical_supplement.py` and its provenance ledger record source hashes, row counts, measured-KL joins, omissions and condition summaries. Original `paper/` sources are unchanged; these six vector figures introduce no model runs, new labels or significance tests.

B Evaluation and Analysis Details

B.1 Elo Estimation and Conditional Uncertainty

For each evaluator and arena, we fit Bradley–Terry ratings from its stored pairwise winner/loser records. With checkpoint ratings R_i, the fitted win probability is

$$p_R(i \succ j) = \sigma\left(\frac{R_i - R_j}{s}\right), \qquad s = \frac{400}{\log 10}.$$

The anchored log-likelihood sums $\log p_R(i \succ j)$ over observed wins, with one reference rating fixed at zero. Each evaluator and contestant set has a separate fit. Additive anchors do not affect within-arena differences, but ratings from different arenas cannot simply be subtracted. In particular, the diagonal of a proxy–Gold plot marks equal numerical ratings, not identical preferences or calibrated agreement.

For the reported uncertainty analysis, we independently resample stored winner/loser rows and refit the ratings 200 times. The refits add an L2 penalty $\frac{0.01}{2}\|\theta_{-s_0}\|^2$ in logit coordinates $\theta = R/s$, with reference s_0 fixed. They are therefore closely related to, but not byte-identical with, the original exported ratings. For configuration c we report

$$\mathrm{Peak}(c) = \max_t R_{c,t}, \qquad \mathrm{Top5}(c) = \frac{1}{5}\sum_{t\in\mathcal{T}_5(c)} R_{c,t},$$

where $\mathcal{T}_5(c)$ contains that run's five highest-rated snapshots. Each resample repeats this selection before computing Answer-Only-minus-Thinking gaps; the reported 95% intervals are bootstrap percentiles.

These intervals describe sensitivity to the stored match table under an independent-row approximation. Comparisons sharing prompts or responses, or edges broken from one partial ranking, can be dependent. The retained winner/loser table does not preserve the cluster identifiers needed for a prompt-aware bootstrap. The intervals do not include fresh generations, independent training seeds, judge changes or tournament redesign. Peak and top-five selection use the evaluation data and should not be read as prospective checkpoint-selection guarantees.

B.2 Descriptive Scaling Fits

We summarize the stored sweep with post-hoc fits, not a theory identifying neural optimization with a cardinal-utility model. For policy size P_c, rewarder size R_c (billions of parameters) and Thinking indicator M_c, ordinary least squares fits

$$Y_c = b_0 + b_P \log P_c + b_R \log R_c + b_M M_c + b_{MP} M_c \log P_c + b_{MR} M_c \log R_c + \epsilon_c,$$

with Y_c the peak or top-five Gold Elo. There are 16 matched mode/size configurations and six fitted coefficients. The interaction terms permit different size slopes for the two pipelines; they do not isolate response mode from its training objective.

Metric	R^2	RMSE	Ans. policy	Ans. judge	Think policy	Think judge
Peak Elo	0.954	51.8	186.0	144.7	113.3	113.3
Top-5 Elo	0.952	51.2	175.8	135.7	119.0	115.6

Table 5: In-sample log-size fits to the 16 matched global configurations. Slope columns are fitted Gold-Elo differences per parameter doubling along each axis, not observed new doublings, compute-matched gains or out-of-sample predictions. RMSE is in Elo units.

Both policy and judge scale have substantial fitted associations with Gold Elo in this grid. The high in-sample R^2 does not establish a universal scaling law: the grid is sparse, checkpoints were selected by the same learned evaluator and no independent-seed uncertainty is included.

For the 16 local KL trajectories, two three-parameter curve families summarize Gold Elo against the square root of the nonnegative KL estimate. Writing $d = \sqrt{\max(D_{\mathrm{KL}}, 0)}$, they are

$$G_c(d) = \gamma_c + a_c d - b_c d^2, \qquad G_c(d) = \gamma_c + d(\alpha_c - \kappa_c \log d),$$

taking $d \log d = 0$ at zero. These are descriptive low-dimensional summaries, following the use of smooth KL curves in reward-overoptimization studies [Gao et al., 2022, Rafailov et al., 2024]. The intercept absorbs each local Elo anchor. Small negative empirical KL estimates are mapped to zero only for this transformation, not interpreted as negative true KL.

Median in-sample RMSE is 63.3 Elo for the quadratic and 63.5 for the logarithmic comparison. Only three quadratic peaks and two logarithmic peaks lie inside the observed KL ranges. Most trajectories therefore do not identify a complete overoptimization frontier; extrapolated stationary points are not deployment recommendations. Coefficients, inputs and analysis scripts are preserved with the research artifacts.

B.3 Why Static Accuracy Is Not an Online-Update Guarantee

A short update-level identity helps interpret the empirical distinction. Let Δ_θ denote the log-probability margin, relative to the reference policy, for a fixed Gold-oriented response pair. The Gold-label loss is $\ell_+ = -\log\sigma(\beta\Delta_\theta)$; a flipped label gives $\ell_- = -\log\sigma(-\beta\Delta_\theta)$. Since $\sigma(u) + \sigma(-u) = 1$,

$$\nabla_\theta \ell_- - \nabla_\theta \ell_+ = \beta \nabla_\theta \Delta_\theta.$$

Thus two rewarders with the same average error can induce different updates if their mistakes occur on pairs with different gradient directions or magnitudes. The relevant pairs are sampled from the evolving policy, not the fixed static validation distribution.

Filtering adds a second difference. The actual trainer sums order-normalized verdicts and retains nonzero-sum pairs; malformed votes contribute zero. Different proxies can therefore change both the labels of retained pairs and the distribution of pairs that survive. High static accuracy alone controls neither effect. This is a possible explanation of the evaluator–rewarder gap, not a measured decomposition: the retained artifacts do not supply the per-pair margins and gradient features needed to estimate it, nor establish which judge's errors better track human preferences.

B.4 Follow-Up Audit with a Different Learned Judge

We report an existing follow-up diagnostic separately from the main Gold arena. For 4B policies trained with 4B and 14B rewarders, we selected each mode's peak-Gold snapshot and sampled 40 validation prompts per setting with seed 17. Response presentation order was randomized. The judge, `Qwen3-235B-A22B-Instruct-2507`, received the prompt and two responses without their training-mode labels. It judged prompt-following, coherence, specificity, originality and writing quality, with an instruction not to favor length. It produced one Answer-Only verdict per pair at temperature zero with a 24-token limit; no swapped-order repeat was performed. All 80 verdicts were parseable.

Rewarder size (4B policy)	Answer-Only wins	Comparisons	95% Wilson interval
4B	17	40	28.5–57.8%
14B	25	40	47.0–75.8%
Combined	42	80	41.7–63.1%

Table 6: A different learned judge does not establish an overall preference advantage in this limited diagnostic. These descriptive binomial intervals are not prompt-cluster or retraining intervals; the 80 comparisons contain 76 unique prompts, with four prompts shared across settings.

The diagnostic is inconclusive overall. Its two settings, selected checkpoints, changed rubric, shared Qwen3 family and single verdict per pair limit interpretation. It neither reproduces a human evaluation nor supplies a thinking-style Gold control; it makes the evaluator dependence of the main result explicit.

B.5 Preliminary Cross-Domain Evaluation: OpenOrca

We retain a limited cross-domain diagnostic on OpenOrca using off-the-shelf Qwen3 policies, judges and Gold evaluator. No fine-tuning or online policy optimization was performed. The plotted proxy–Gold net-score relationship is a diagnostic of these off-the-shelf evaluators; it does not replicate the trained creative-writing result or establish response-level verdict agreement. Generalization of the evaluator–rewarder gap to another domain remains untested.

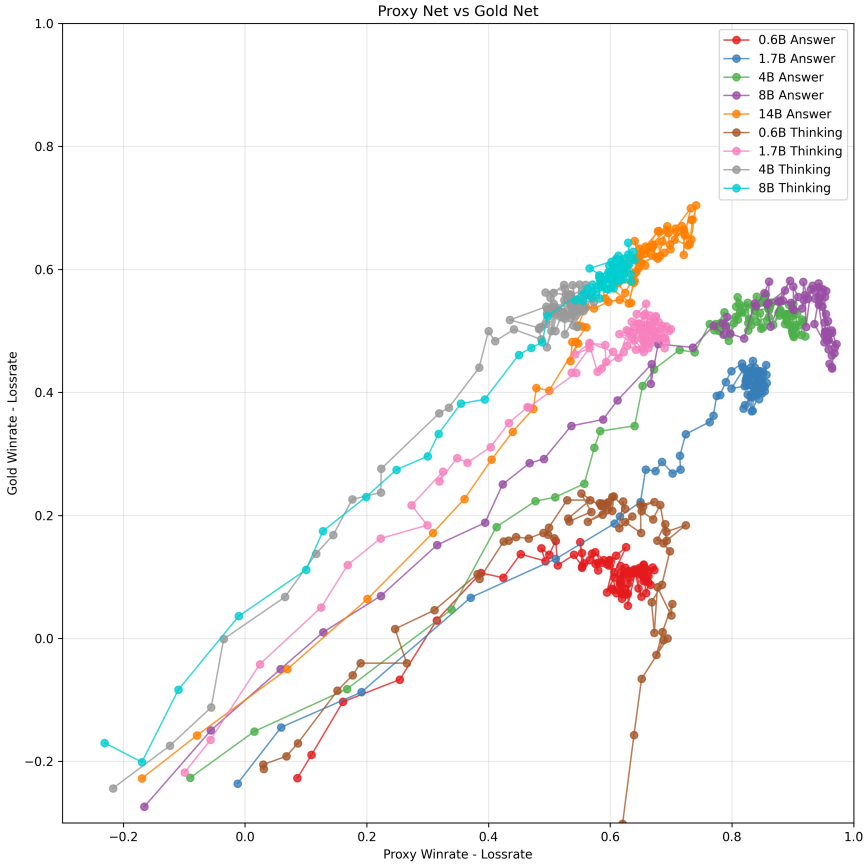


Figure 13: Proxy vs. Gold net score on OpenOrca. All models are off-the-shelf (no GenRM training). A full replication with trained GenRMs and policy optimization remains future work.

C Prompt Templates

We provide the exact prompt templates used for policy generation and GenRM pairwise evaluation. All models receive these prompts without modification.

C.1 Policy Prompt

The policy model receives the following prompt template for creative writing generation:

```
Creative Writing Prompt:
{QUESTION}
Goal: write an original short story that directly answers the prompt.
Requirements: Output only the story text. no titles, notes, commentary
or meta-text.
```

The `QUESTION` placeholder is replaced with the specific creative writing instruction from the evaluation set.

C.2 GenRM Pairwise Evaluation Prompt

The generative reward model receives the following prompt template for pairwise comparison:

```
Two AI models were given the same instruction and each produced a reply.
Your task is to judge which reply better fulfills the instruction.

[BEGIN INSTRUCTIONS GIVEN TO BOTH MODELS]
{POLICY_PROMPT}
[END INSTRUCTIONS]

[BEGIN REPLY A]
{RESPONSE_A}
[END REPLY A]

[BEGIN REPLY B]
{RESPONSE_B}
[END REPLY B]

Output ONLY one of:
<answer>[[A]]</answer>
<answer>[[B]]</answer>
```

The placeholders are filled with the full policy prompt and the two candidate responses. The model outputs a structured verdict indicating which response better satisfies the original instruction.